

	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	1 of 21

PROJECT **MPA RE-ACTIVATION PROJECT**

CLIENT **PT.MULTI OCEAN SHIPYARD**

CUSTOMER **PT.GERBANG SARANABAJA**

ITEM NO. **S-3010 & S-3000**

SERVICE **PIG LAUNCHER & PIG RECEIVER**

DESCRIPTION **QUICK OPENING CLOSURE**

DOCUMENT No. **Q24-102-STR-01**

0	2024/12/11	ISSUED FOR APPROVAL	J. E. T	T. K. LIM	D. Y. NOH
Rev.	DATE	DESCRIPTION	PREPARED BY	REVIEWED BY	APPROVED BY

	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	2 of 21

1. Definition	-----	3 page
2. Design Condition	-----	4 page
2-1 Design Code		
2-2 Design Data		
2-3 Materials		
2-4 Allowable Stress		
2-5 Elasticity Modulus		
3. Analysis Condition	-----	6 page
3-1 Contact Definition		
3-2 Mesh of Structure		
3-3 Finite Element Model		
3-4 Geometry Model, Load and Boundary Condition		
4. Deformation Result	-----	10 page
5. Equivalent Stress of Structure	-----	11 page
6. Stress Analysis for Hub	-----	12 page
6-1 Equivalent (von-Mises) Stress		
6-2 Linearized Equivalent Stress (Path-1)		
6-3 Linearized Equivalent Stress (Path-2)		
7. Stress Analysis for Door	-----	15 page
7-1 Equivalent (von-Mises) Stress		
7-2 Linearized Equivalent Stress (Path-3)		
7-3 Linearized Equivalent Stress (Path-4)		
8. Stress Analysis for Locking Band	-----	18 page
8-1 Equivalent (von-Mises) Stress		
8-2 Linearized Equivalent Stress (Path-5)		
8-3 Linearized Equivalent Stress (Path-6)		
9. Conclusion	-----	21 page

	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	3 of 21

1. Definition

- Frictionless : This setting models standard unilateral contact; that is, normal pressure equals zero if separation occurs. Thus gaps can form in the model between bodies depending on the loading. This solution is nonlinear because the area of contact may change as the load is applied. A zero coefficient of friction is assumed, thus allowing free sliding. The model should be well constrained when using this contact setting. Weak springs are added to the assembly to help stabilize the model in order to achieve a reasonable solution.
 - No Separation : This contact setting is similar to the Bonded case. It only applies to regions of faces (for 3D solids) or edges (for 2D plates). Separation of the geometries in contact is not allowed.
 - Displacement : Displacements are applied at the geometry level. They require that one or more flat or curved faces or edges or one or more vertices to displace relative to their original location by one or more components of a displacement vector in the world coordinate system or local coordinate system, if applied.
 - Equivalent stress : Equivalent stress (also called von Mises stress) is often used in design work because it allows any arbitrary three-dimensional stress state to be represented as a single positive stress value. Equivalent stress is part of the maximum equivalent stress failure theory used to predict yielding in a ductile material.
 - Membrane Stress : Average stress value across the thickness of a solid section, excluding the effects of discontinuities and stress concentrations.
 - Bending Stress : Portion of primary stress proportional to the distance from the centroid of a cross-section, excluding the effects of discontinuities and stress concentrations.
 - Deformation : Physical deformations can be calculated on and inside a part or an assembly. Fixed supports prevent deformation; locations without a fixed support usually experience deformation relative to the original location. Deformations are calculated relative to the part or assembly world coordinate system.
-

	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	4 of 21

2. Design Condition

2-1 Design Code

Application Code : ASME SEC. VIII DIV.1 2023 ED
Applied Ansys Version (Workbench and Mechanical APDL) : R17.0

2-2 Design Data

Vessel Orientation	:	Horizontal
Contents	:	-
Description	:	Quick Opening Closure
Design Pressure	:	9.31 MPa(g) (93.08 bar(g))
Design Temperature	:	93.3 °C
Min. Design Metal Temperature	:	0.0 °C
Vessel Max. Allowable Working Pressure	:	9.37 MPa(g) (93.7 bar(g))
Joint Efficiency (Hub / Door)	:	1.0 / 1.0 (Seamless)
Corrosion Allowance	:	3.2 mm
Vessel Hydrostatic Test Pressure	:	12.18 MPa(g) (121.81 bar(g))
Poisson Ratio	:	0.3

	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	5 of 21

2-3 Material (ASME Section II part A)

Hub Material	:	SA105
Door Material	:	SA105
Band Material	:	316L S.S

2-4 Allowable Stress for Design Condition (ASME Section II part D Table 1A)

Hub	=	138.0	MPa	At Design Temperature
Door	=	138.0	MPa	At Design Temperature
Band	=	115.0	MPa	At Design Temperature

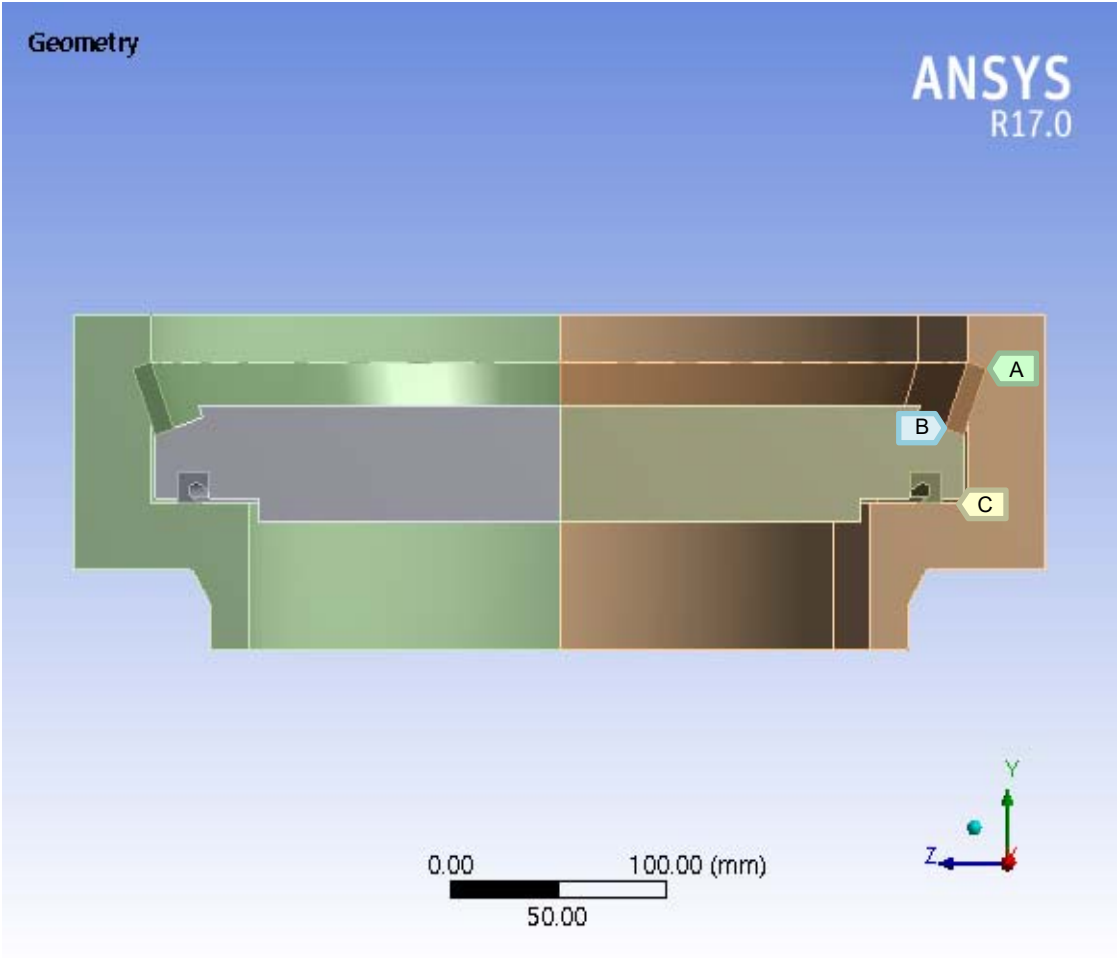
2-5 Elasticity Modulus for Design Condition (ASME Section II part D Table TM-1)

Hub	=	198356.0	MPa	At Design Temperature
Door	=	198356.0	MPa	At Design Temperature
Band	=	189533.0	MPa	At Design Temperature

	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	6 of 21

3. Analysis Condition

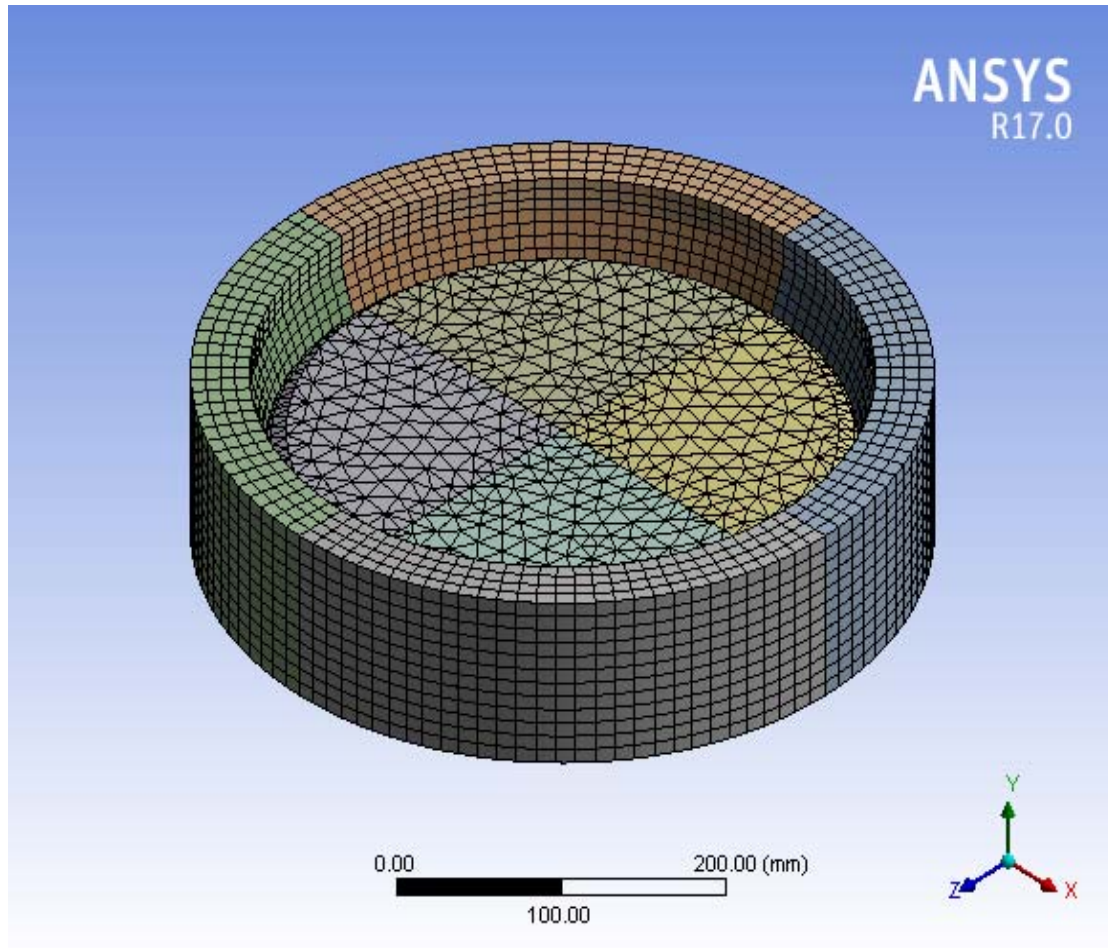
3-1 Contact Definition



Description	Geometry	Contact
A	Hub to Band	No Separation
B	Door to Band	No Separation
C	Hub to Door	Frictionless

	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	7 of 21

3-2 Mesh of Structure



- ▷ PLANE 185 is used for 3-D modeling of solid structures.

JCONJIN ENTECH LTD.	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	8 of 21

3-3 Finite Element Model

Finite element model (SOLID185) is prepared for the analysis. Following element is used for the analysis.

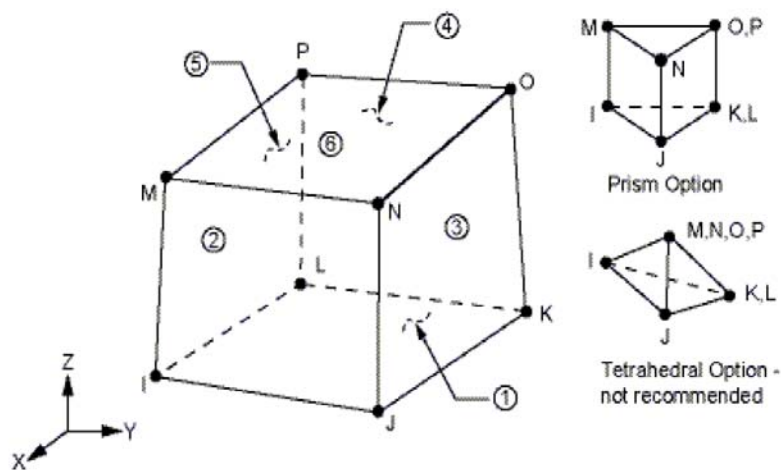
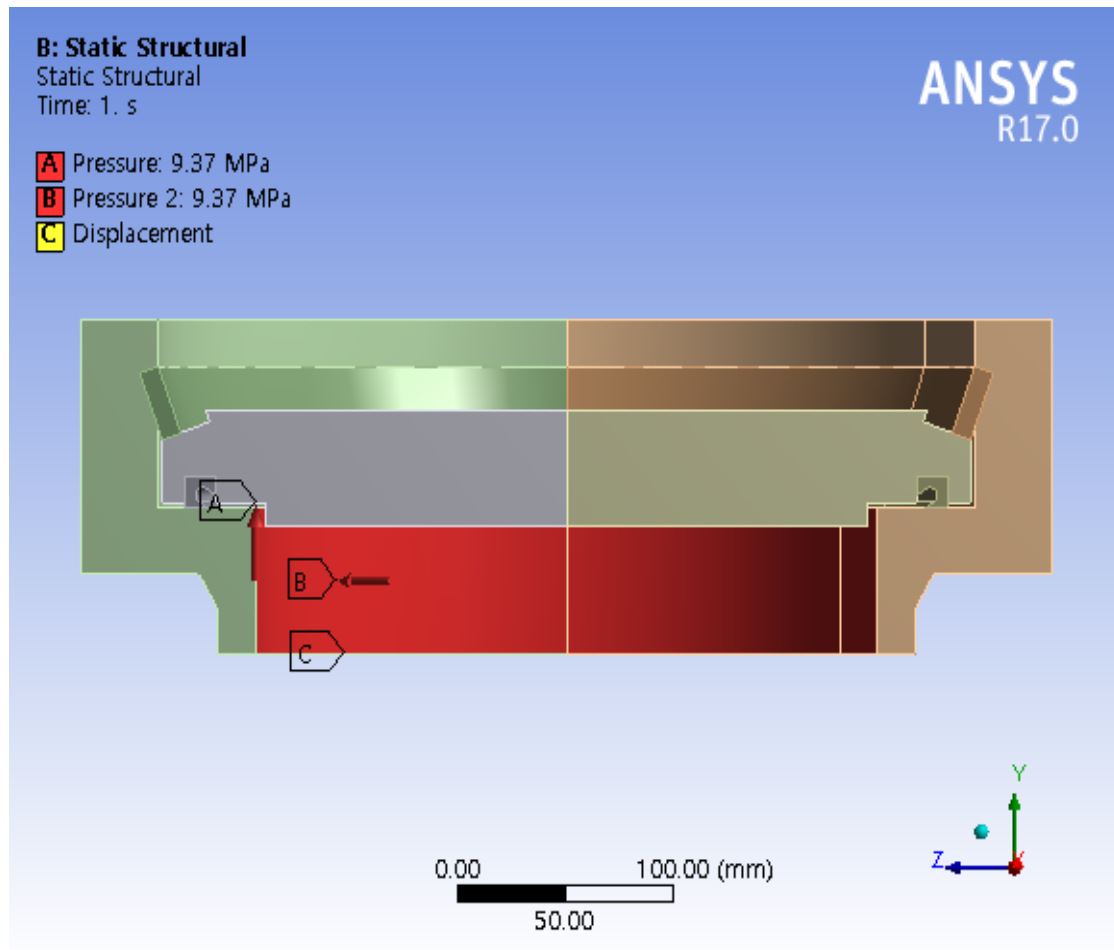


Figure 4.2 SOLID Geometry

SOLID185 is used for 3-D modeling of solid structures. It is defined by eight nodes having three degrees of freedom at each node: translations in the nodal X, Y, and Z directions. The element has plasticity, hyperelastic stress stiffening, creep, large deflection, and large strain capabilities. It also has mixed formulation capability for simulating deformations of nearly incompressible hyperelastic materials.

	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	9 of 21

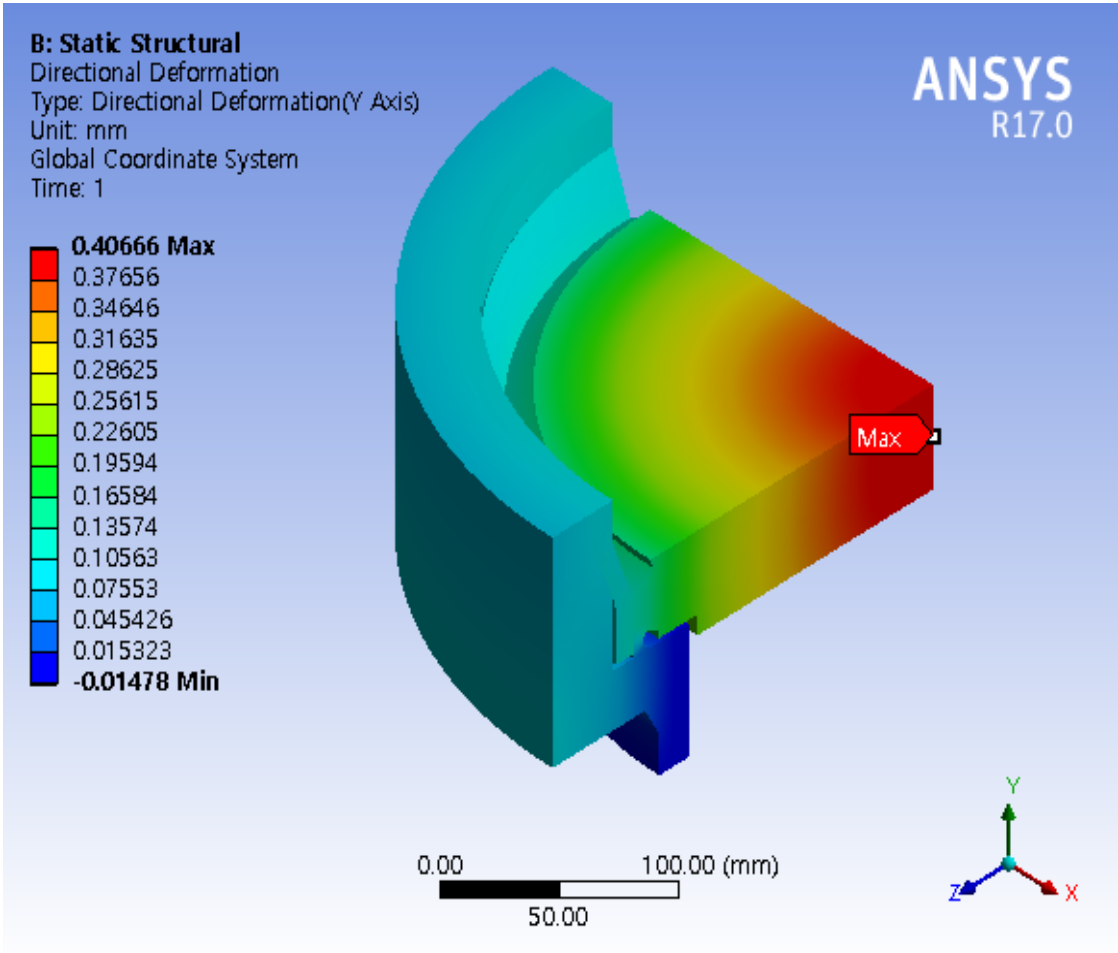
3-4 Geometry Model, Load and Boundary Condition



- ▷ Geometry : Hub, Door and Locking Band
- ▷ Displacement : UY = 0

	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	10 of 21

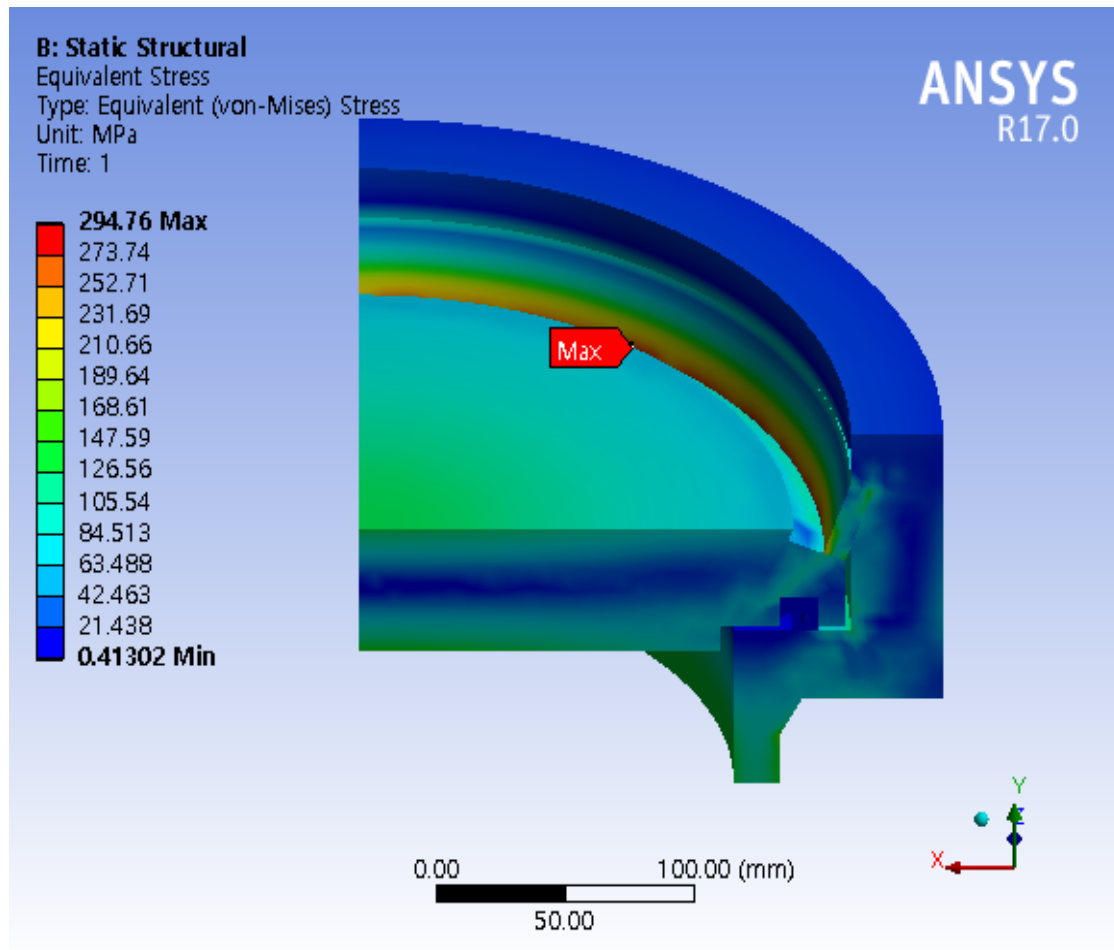
4. Deformation Result in Y-direction



▷ Maximum Deformation Value : 0.41 mm (Y-direction)

	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	11 of 21

5. Equivalent Stress of Structure

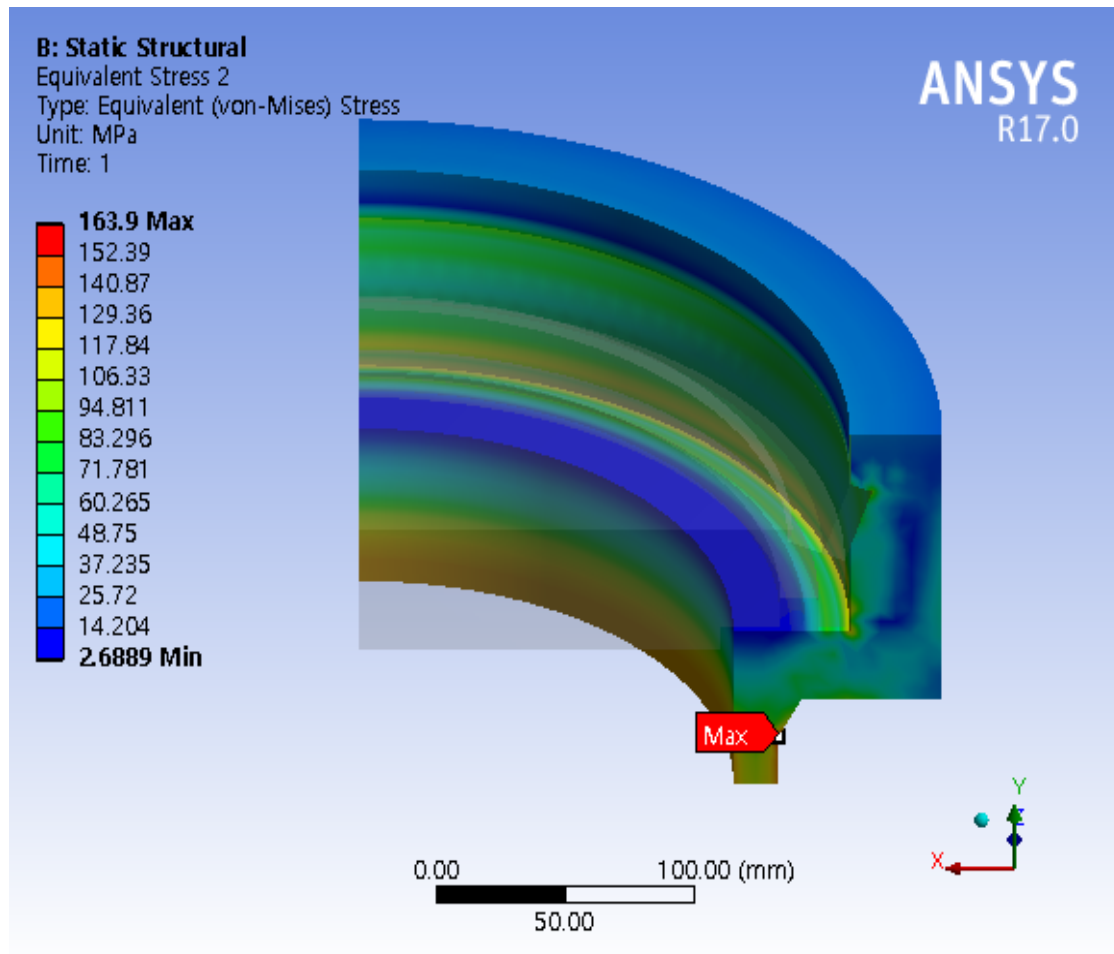


▷ Refer to equivalent stress result of each parts on continued page

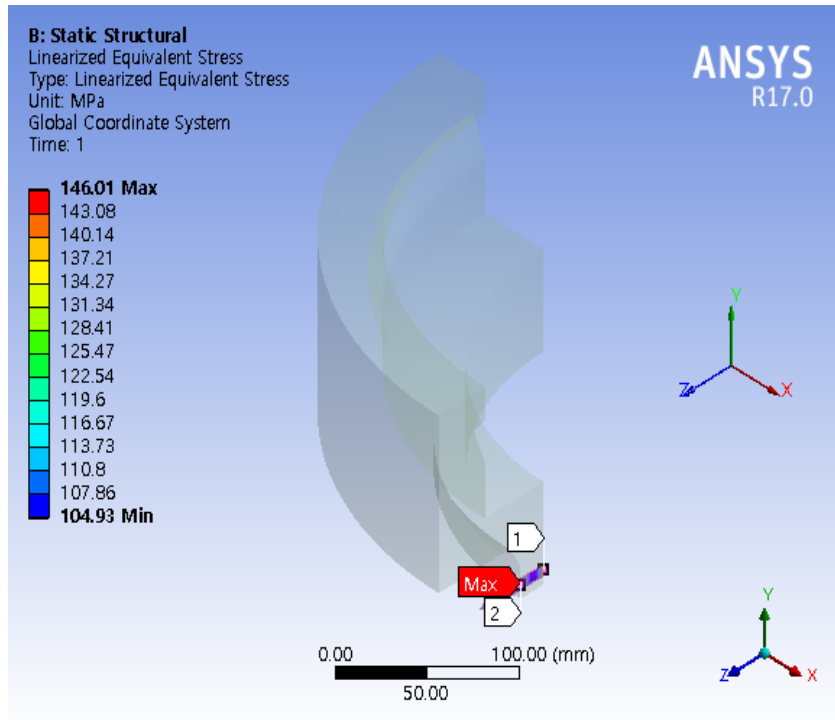
	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	12 of 21

6. Stress Analysis for Hub

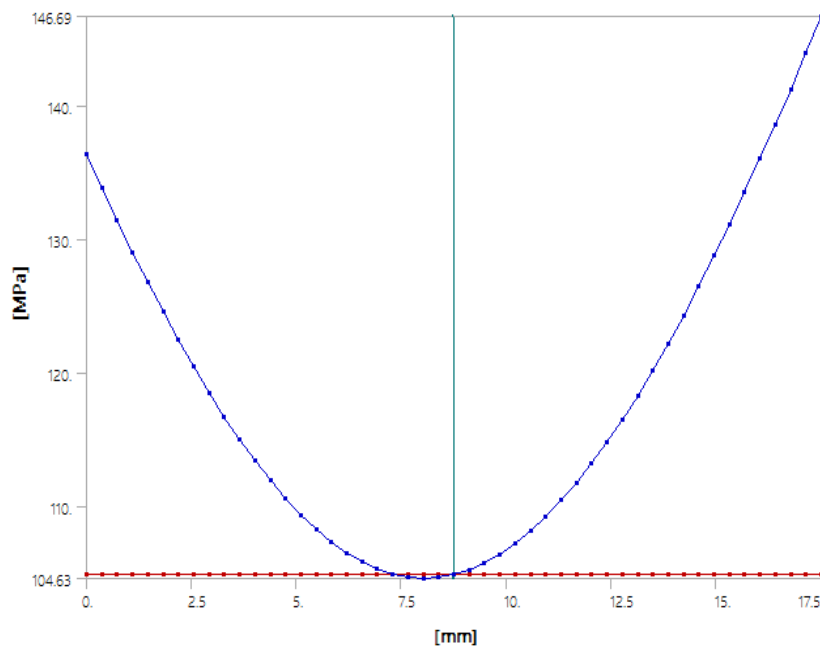
6-1 Equivalent (von-Mises) Stress



6-2 Linearized Equivalent Stress (Path-1)

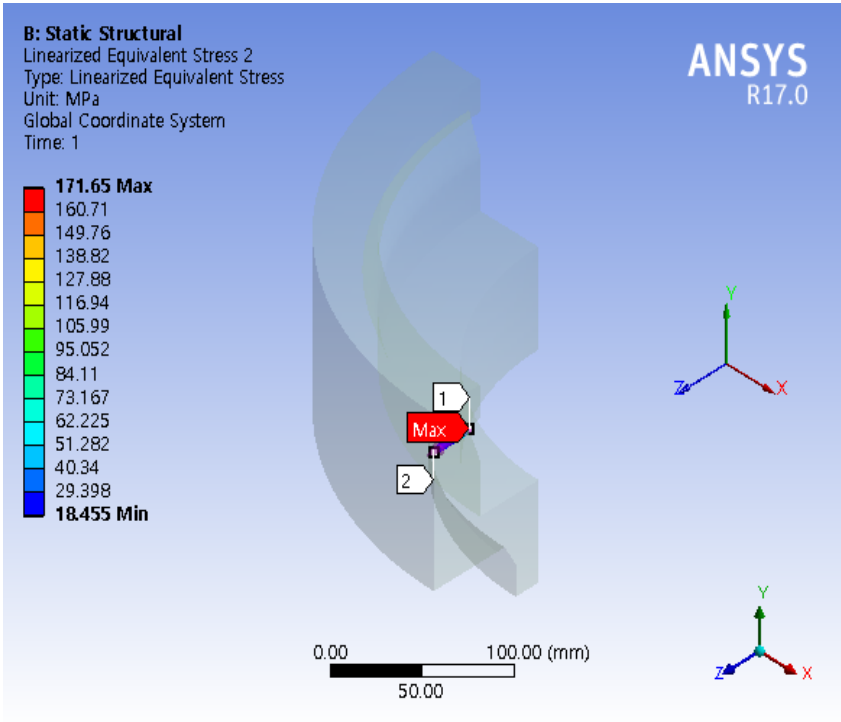


- ▷ **Pm = 104.91 Mpa** (Pm : Membrane)
- ▷ **Pm+Pb = 146.69 Mpa** (Pm+Pb : Membrane + Bending)

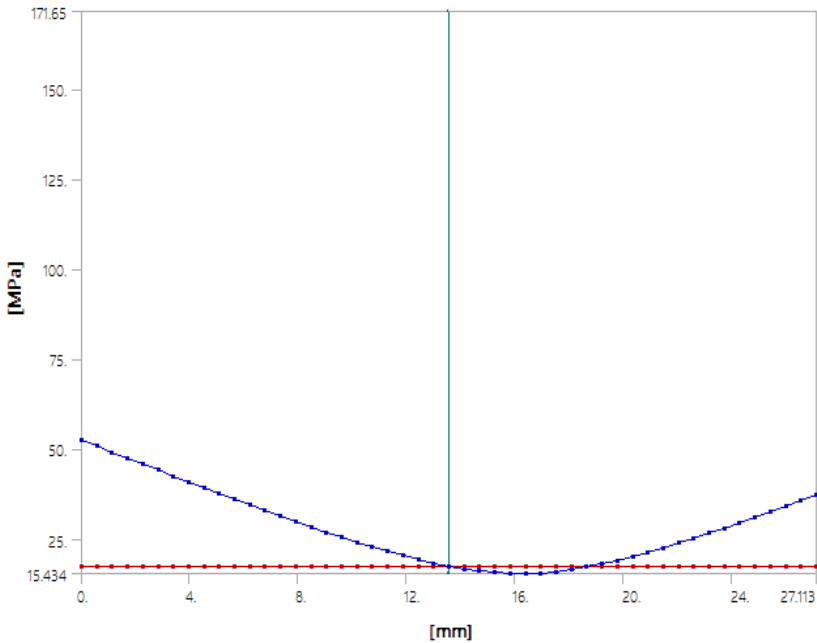


	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	14 of 21

6-3 Linearized Equivalent Stress (Path-2)

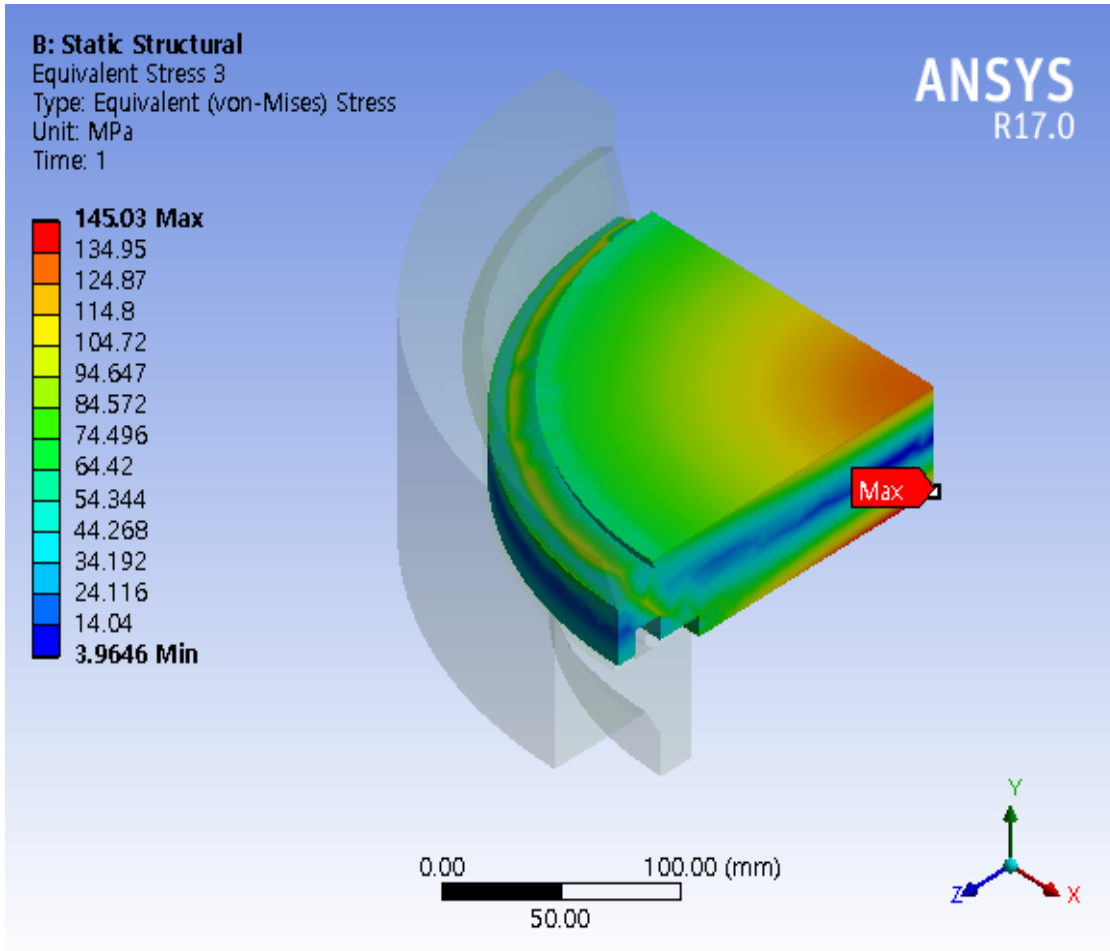


- ▷ **Pm = 17.40 Mpa** (Pm : Membrane)
- ▷ **Pm+Pb = 52.49 Mpa** (Pm+Pb : Membrane + Bending)

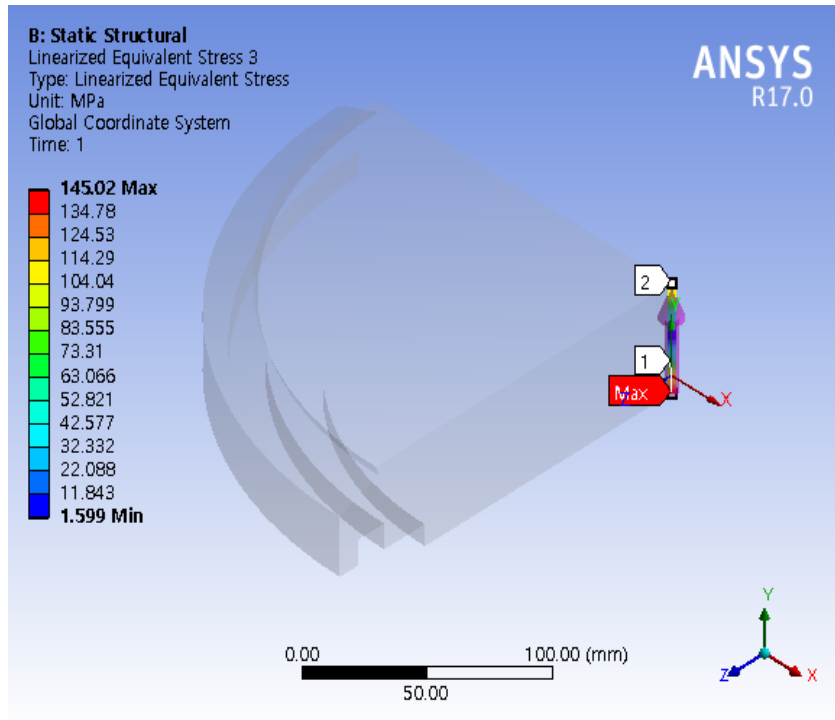


7. Stress Analysis for Door

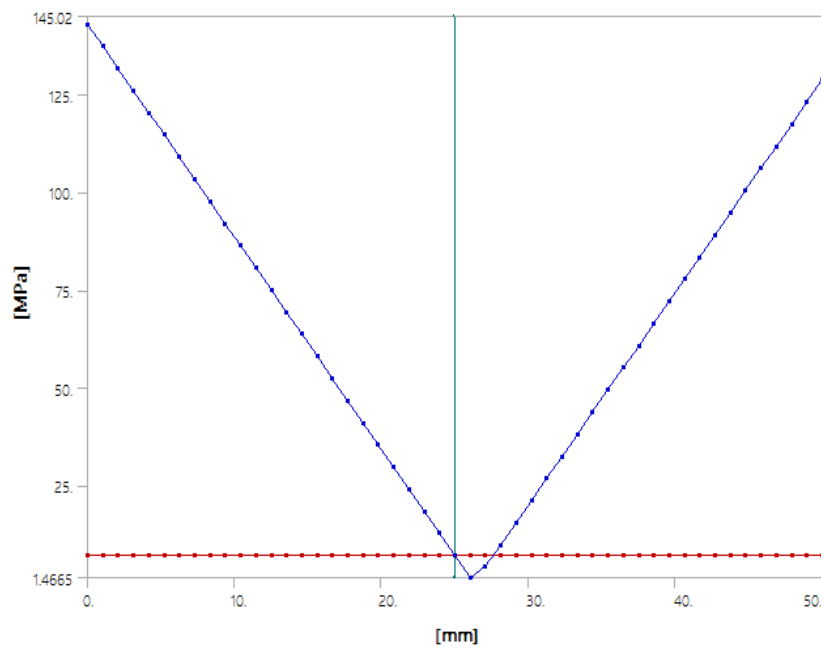
7-1 Equivalent (von-Mises) Stress



7-2 Linearized Equivalent Stress (Path-3)

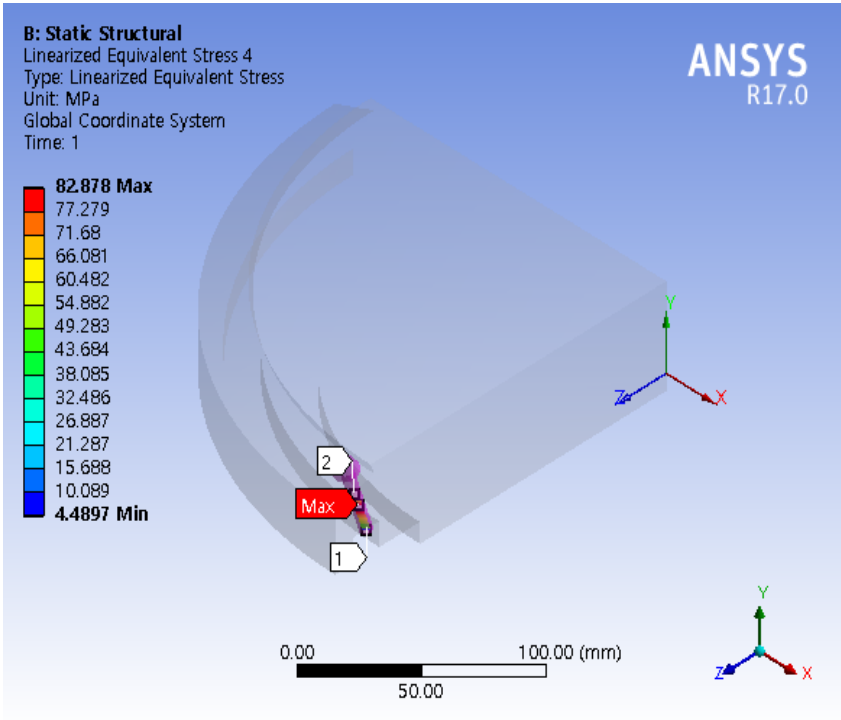


- ▷ **Pm = 7.13 Mpa** (Pm : Membrane)
- ▷ **Pm+Pb = 143.02 Mpa** (Pm+Pb : Membrane + Bending)



	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	17 of 21

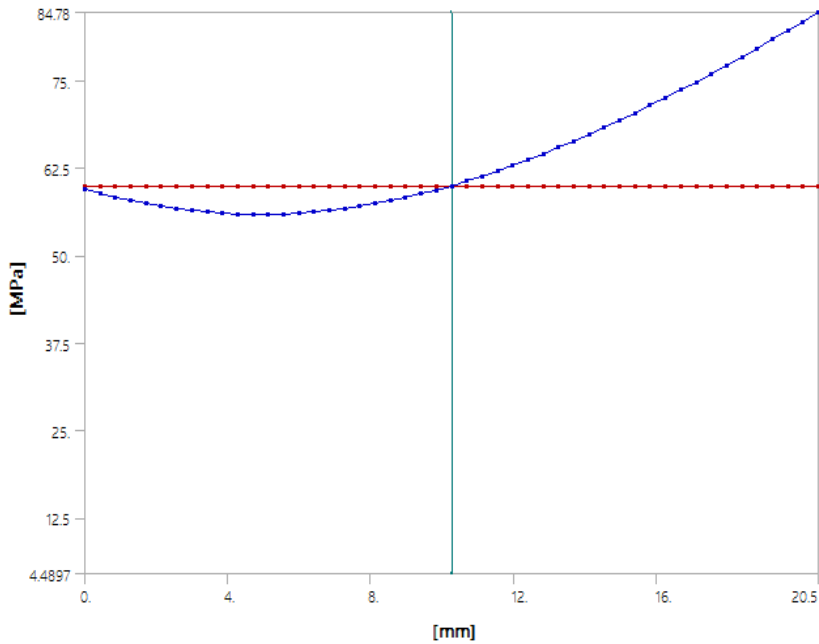
7-3 Linearized Equivalent Stress (Path-4)



- ▷

Pm =
59.97
Mpa
 (Pm : Membrane)
- ▷

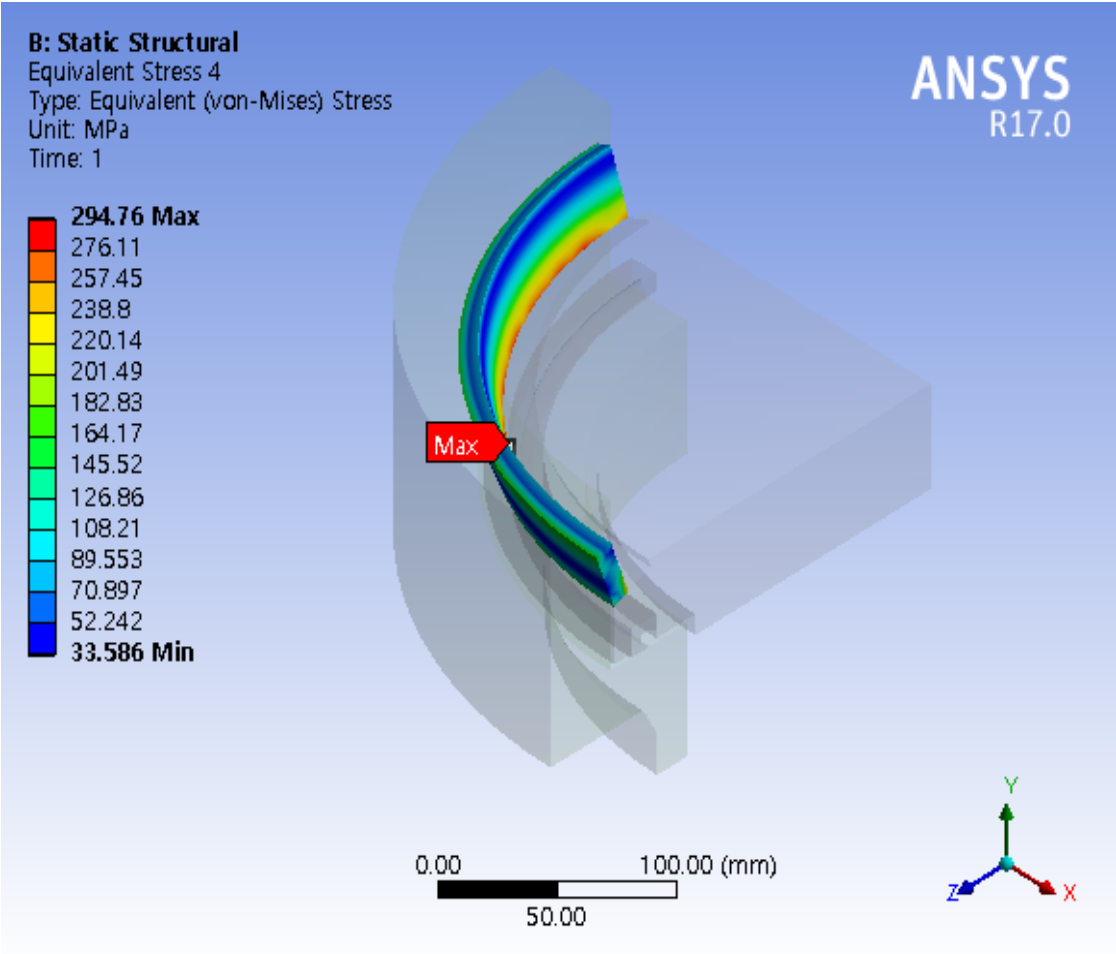
Pm+Pb =
84.78
Mpa
 (Pm+Pb : Membrane + Bending)



	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	18 of 21

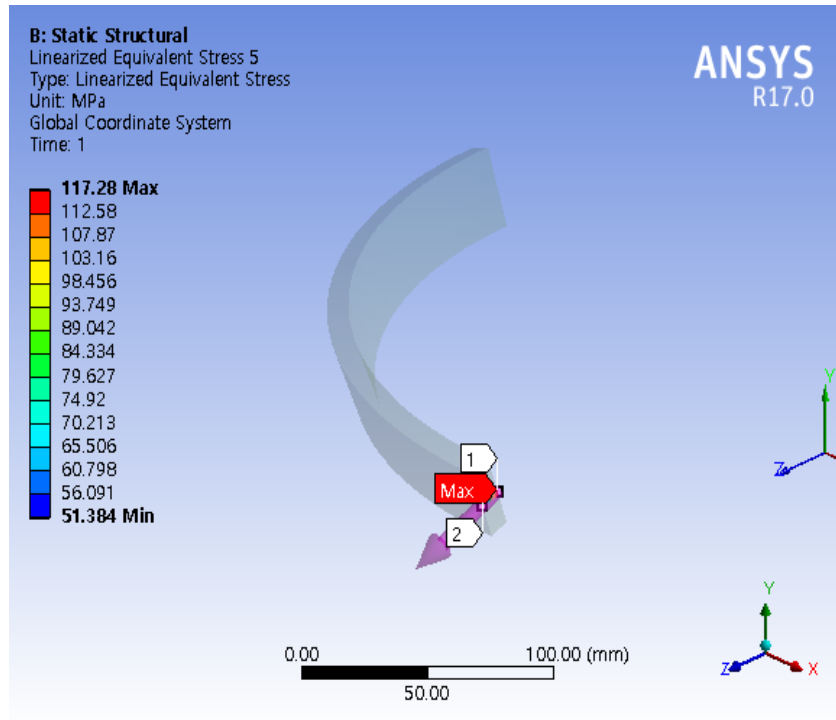
8. Stress Analysis for Locking Band

8-1 Equivalent (von-Mises) Stress

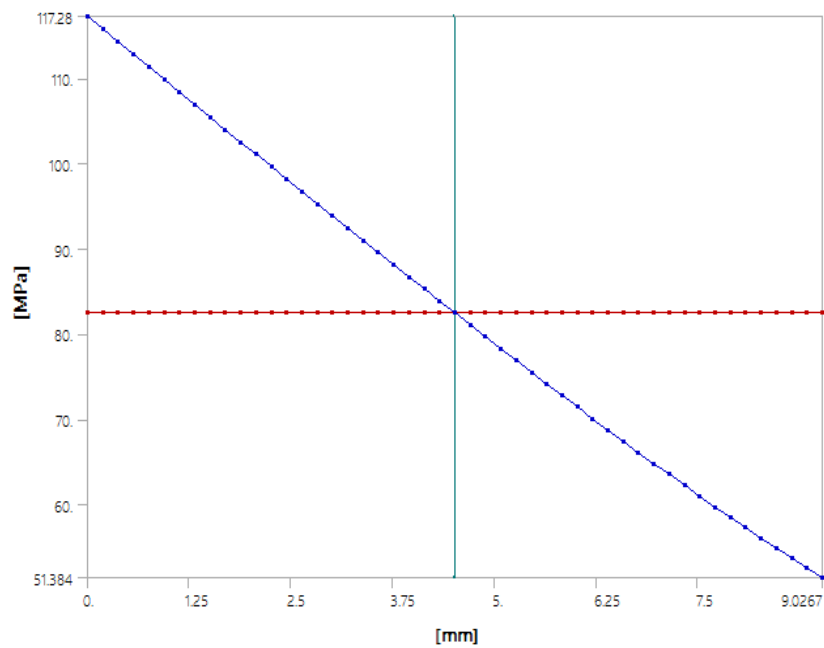


	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	19 of 21

8-2 Linearized Equivalent Stress (Path-5)

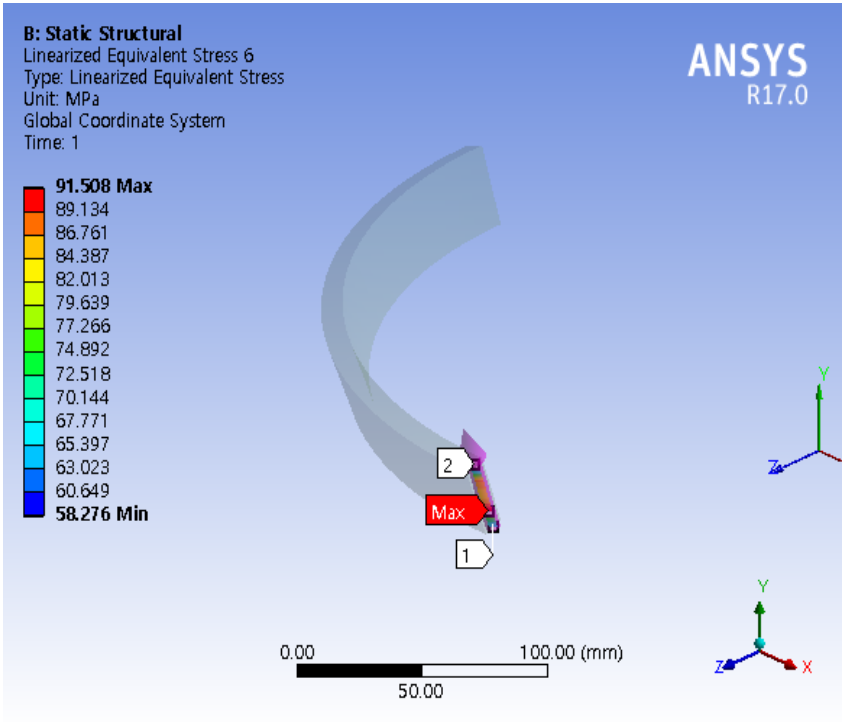


- ▷ **Pm = 82.44 Mpa** (Pm : Membrane)
- ▷ **Pm+Pb = 117.27 Mpa** (Pm+Pb : Membrane + Bending)

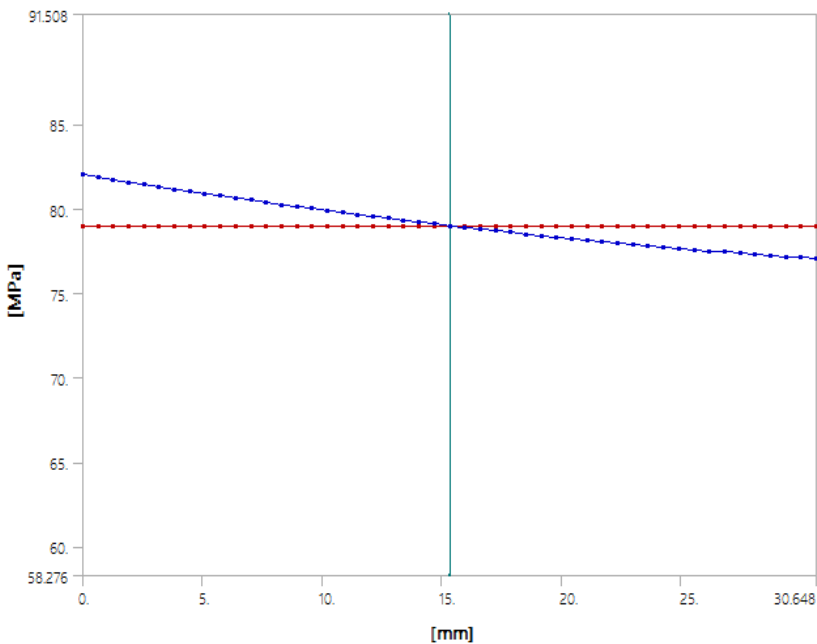


	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	20 of 21

8-3 Linearized Equivalent Stress (Path-6)



- ▷ **Pm = 78.99 Mpa** (Pm : Membrane)
- ▷ **Pm+Pb = 82.01 Mpa** (Pm+Pb : Membrane + Bending)



	STRUCTURAL ANALYSIS	DOC. NO.	Q24-102-STR-01
		REV. NO.	0
		SHEET	21 of 21

9. Conclusion

The evaluation is carried out according to ASME Sec. VIII Div. 2.

The stress on evaluation points is calculated and evaluated.

As a result, membrane stress and membrane plus bending stress are shown on Table 9-1.

√ General Primary Membrane Equivalent Stress (P_m) < S_a

√ General Primary Membrane plus Bending Equivalent Stress ($P_m + P_b$) < $1.5 \cdot S_a$

Table 9-1 Result Summary of Stress calculation at design condition (Unit = Mpa)

Component	P_m	Allowable < $1.0S_a$	$P_m + P_b$	Allowable < $1.5S_a$	Result
Hub : Path1	104.91	138.0	146.69	207.0	Passed
Hub : Path2	17.40	138.0	52.49	207.0	Passed
Door : Path3	7.13	138.0	143.02	207.0	Passed
Door : Path4	59.97	138.0	84.78	207.0	Passed
Band : Path5	82.44	115.0	117.27	172.5	Passed
Band : Path6	78.99	115.0	82.01	172.5	Passed